

Task 2: Vulnerability Scanner (Mini Project)

Objective:

Build a simple Python-based vulnerability scanner that detects common security issues in a web application or network.

Project Overview

A Vulnerability Scanner is a tool used to identify security weaknesses in systems, websites, or networks. This mini-project focuses on:

- Checking open ports
 - Detecting weak configurations
 - Identifying outdated software versions
 - Generating a vulnerability report
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Technologies Used

- Python 3
- Socket Library
- Requests Library
- Nmap (optional)
- OS Module

Install required packages:

```
pip install requests python-nmap
```

Project Structure

Vulnerability-Scanner/

```
|
|— scanner.py
|— report.txt
|— requirements.txt
|— README.md
```

Features

1. Port Scanning

Checks commonly used ports:

- 21 (FTP)
 - 22 (SSH)
 - 80 (HTTP)
 - 443 (HTTPS)
 - 3306 (MySQL)
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2. HTTP Security Header Check

Detects missing headers:

- X-Frame-Options
 - Content-Security-Policy
 - X-XSS-Protection
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3. Server Information Detection

Obtains server information from HTTP response headers.

4. Report Generation

Creates a text report containing all findings.

scanner.py

```
import socket
import requests
```

```
# -----
# Port Scanner
```

```

# -----
def scan_ports(target):
    print(f"\nScanning Ports on {target}\n")

    ports = [21, 22, 80, 443, 3306]
    open_ports = []

    for port in ports:
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        s.settimeout(1)

        result = s.connect_ex((target, port))

        if result == 0:
            print(f"Port {port} is OPEN")
            open_ports.append(port)

        s.close()

    return open_ports

# -----
# Security Header Checker
# -----
def check_headers(url):
    print("\nChecking Security Headers...\n")

    headers_found = {}

    try:
        response = requests.get(url, timeout=5)

        security_headers = [
            "X-Frame-Options",
            "Content-Security-Policy",
            "X-XSS-Protection"
        ]

        for header in security_headers:
            headers_found[header] = response.headers.get(header, "Missing")

    except Exception as e:
        print("Error:", e)

```

```

return headers_found

# -----
# Report Generator
# -----
def generate_report(target, ports, headers):
    with open("report.txt", "w") as file:

        file.write("VULNERABILITY SCAN REPORT\n")
        file.write("=" * 40 + "\n\n")

        file.write(f"Target: {target}\n\n")

        file.write("Open Ports:\n")
        for port in ports:
            file.write(f"- {port}\n")

        file.write("\nSecurity Headers:\n")

        for key, value in headers.items():
            file.write(f"{key}: {value}\n")

    print("\nReport Generated Successfully!")

# -----
# Main Program
# -----
target = input("Enter IP Address: ")
url = input("Enter Website URL: ")

ports = scan_ports(target)

headers = check_headers(url)

generate_report(target, ports, headers)

```

Sample Output

```

Enter IP Address: 142.250.183.206
Enter Website URL: https://google.com

```

Scanning Ports on 142.250.183.206

Port 80 is OPEN

Port 443 is OPEN

Checking Security Headers...

Report Generated Successfully!

Generated Report (report.txt)

VULNERABILITY SCAN REPORT

=====

Target: 142.250.183.206

Open Ports:

- 80

- 443

Security Headers:

X-Frame-Options: SAMEORIGIN

Content-Security-Policy: Missing

X-XSS-Protection: 0

How to Run

1. Save the code as `scanner.py`
2. Open terminal in project folder
3. Run:

python [scanner.py](#)

4. Enter target IP and website URL.
 5. View generated report in `report.txt`.
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